Opus 4.8

```
## Setting Up Coordinates
Let me place the points on a
  number line with $A = 0$.
**Finding positions from the
  given distances:**
- $AC = 26 \Rightarrow C =
  26$
- $AF = 73 \Rightarrow F =
  73$
- $CE = 31 \Rightarrow E =
  57$
- $DF = 33 \Rightarrow D =
  73 - 33 = 40$
- $BD = 22 \Rightarrow B =
  40 - 22 = 18$
```

Kimi-K2.6 (prefilled)

```
Place the six collinear
  points on the $x$-axis.
Set $A$ at the origin, so
  $a=0$.

From the given lengths:
- $AC = 26 \Rightarrow c =
  26$
- $AF = 73 \Rightarrow f =
  73$
- $CE = 31 \Rightarrow e =
  c+31 = 57$
- $DF = 33 \Rightarrow d =
  f-33 = 40$
- $BD = 22 \Rightarrow b =
  d-22 = 18$

Thus the coordinates are
$$A(0),\; B(18),\; C(26),\;
  D(40),\; E(57),\; F(73).$$
```

Opus 4.6

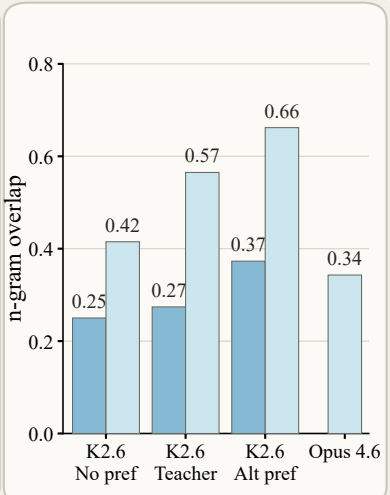
```
## Setting Up Coordinates

I place the points on a
  number line with **A =
  0**. Using the given
  distances:

| Equation | Result |
|---|---|
| AF = 73 | **F = 73** |
| AC = 26 | **C = 26** |
| DF = 33 → d = 73 − 33 |
  **D = 40** |
| BD = 22 → b = 40 − 22 |
  **B = 18** |
| CE = 31 → e = 26 + 31 |
  **E = 57** |

**Verification of order:**
  A(0) < B(18) < C(26) <
  D(40) < E(57) < F(73) ✓
```

(a) Geometry answer excerpts



(b) Geometry overlap

Opus 4.8

```
## Setting Up the Solution
The equation is linear:
$$x_3' + \tanh(t) x_3 = e^{-
  t/3}.$$
**Integrating factor:**
$$\mu(t)=\exp\left(\int\tanh
  t\,dt\right)=\exp(\ln\cosh
  t)=\cosh t.$$
```

Kimi-K2.6 (unprefilled)

```
The given first-order linear
  ODE is
$$x_3'(t)+\tanh(t)\,x_3(t)=e
  ^{-t/3}, \qquad x_3(0)=5.
  $$

**1. Solve the ODE.**
The integrating factor is
$$\mu(t)=\exp\!\left(\int
  \tanh t\,dt\right)=\exp\!
  \bigl(\ln\cosh t\bigr)=
  \cosh t.$$
```

GPT-OSS-120B (unprefilled)

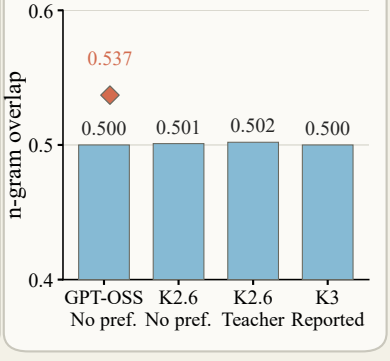
```
The linear equation

\[
x_3'(t)+\tanh t\,x_3(t)=e^{-
  t/3},\qquad x_3(0)=5
\]

has integrating factor

\[
\mu(t)=\exp\!\Bigl(\int
  \tanh t\,dt\Bigr)=\exp(\ln
  \cosh t)=\cosh t .
\]
```

(c) ODE answer excerpts



(d) ODE overlap